F78AB Actuarial and Financial Mathematics B - Solutions to May 2016 Exam

1. (a) The accumulation of £1 from time 0 to time t is

$$A(t) = \exp\left\{\int_0^t (0.04 + 0.001s)ds\right\} = e^{0.04t+0.0005t^2}.$$

For $t = 9$,

$$A(9) = e^{0.4005} = 1.49257$$

(b) The accumulation of £1 from time $t = 8$ to time $t = 9$ is

$$A(9)/A(8) = 1.49257/1.42191 = 1.04969$$

(c) i. The nine-year spot rate of interest y_9 from time $t = 0$ to time $t = 9$

$$(1 + y_9)^9 = 1.49257 \quad \Rightarrow \quad y_9 = 4.550\%$$

ii. The one-year forward rate of interest $f_{8,1}$ from time $t = 8$ is

$$(1 + f_{8,1}) = A(9)/A(8) = 1.04970 \quad \Rightarrow \quad f_{8,1} = 4.970\%$$

2. (a) A forward contract is an agreement between two parties to exchange a fixed quantity of an asset for a fixed price (the forward price) at a fixed time. The long forward position buys the asset for the forward price, and the short forward position sells the asset at the forward price.

(b) The no-arbitrage forward price for the contract when it was agreed three years ago is

$$K' = \left(82 - 6(1.05)^{-7}\right) \times 1.05^{10} = 126.62.$$

The no-arbitrage forward price if the contract had been agreed today would be

$$K = \left(54 - 6(1.05)^{-4}\right) \times 1.05^{7} = 69.04.$$

Thus the value of the contract to the investor (long forward) is

$$\left(K - K'\right) \times 1.05^{-7} = -57.58 \times 1.05^{-7} = -£40.92.$$

Alternatively, could use directly $54 - 82 \times 1.05^3$.

3. (a) The annual discount factor is $v = 1.07^{-1}$. Work in £million.
At 1 January 2017, the present value of the cost of the contract is

$$40 + 50\bar{a}_{\overline{2}|} = 40 + 50\frac{1 - v^2}{\delta} = 40 + 50(1.87059) = 133.529.$$

The present value of the net income is

$$17v^2\bar{a}_{\overline{15}|} = 17v^2(9.42310) = 139.919.$$

The net present value is $139.919 - 133.529 = £6.390$ million.

(b) Let time be measured in years, with time 0 denoting 1 January 2019. At 1 January 2019, the accumulated value of the costs is $133.529 \times 1.07^2 = 152.877$, and the accumulated net income is 0. Let T be the time at which the accumulated value of the net income exceeds the accumulated value of the costs, i.e. T is the first time t at which

$$17\bar{s}_{\overline{t}|} \geq 152.877 \times 1.07^t$$

Since income is received continuously, T will give equality. Rearranging,

$$17\frac{1.07^t - 1}{\ln(1.07)} \geq 152.877 \times 1.07^t \quad \Rightarrow \quad 1.07^t = \left(1 - \frac{152.877\ln(1.07)}{17}\right)^{-1}$$

Taking logs,

$$t = -\ln\left(1 - \frac{152.877\ln(1.07)}{17}\right) / \ln(1.07) = 13.858.$$

The discounted payback period is $2 + 13.858 = 15.858$ years from 1 January 2017.

(c) The discounted payback period would be less than in part (b). As the income is received after the costs are paid, the lower interest rate increases the discounted value of the income more than the discounted value of the costs.

4. (a) Work in £million. The force of interest is $\delta = 0.05$ p.a. and let $v = e^{-0.05}$.

The PV PV_L and duration τ_L of the liability payments are

$$PV_L := 10v^{12} + 20v^{15} = 14.93545$$

$$\tau_L = \frac{12 \times 10v^{12} + 15 \times 20v^{15}}{PV_L} = \frac{207.56736}{PV_L}.$$

Denote by X_5 the amount invested today in the 5-year ZCB, and by X_{25} the amount invested today in the 25-year ZCB. The PV PV_A and duration τ_A of the assets are

$$PV_A := X_5 + X_{25} \qquad \tau_A = \frac{5 \times X_5 + 25 \times X_{25}}{PV_A}.$$

Equating PVs, rearrange to get $X_{25} = 14.93545 - X_5$. Equating durations, and substituting for X_{25}, we find

$$X_5 = \frac{25 \times 14.93545 - 207.56736}{25 - 5} = 8.29094$$

and hence

$$X_{25} = 14.93545 - X_5 = 14.93545 - 8.29094 = 6.64451$$

(b) The force of interest is now $\delta = 0.055$ p.a. The PV of the liabilities is

$$PV_L := 10v^{12} + 20v^{15} = 13.93321$$

The PV of the assets also changes. The redemption payments are

for X_5, $8.29094 \times e^{0.05\times 5} = 10.64578$ and for X_{25}, $6.64451 \times e^{0.05\times 25} = 23.19162$.

With the change in interest rate, the PV of the assets is

$$10.64578e^{-0.055\times 5} + 23.19162e^{-0.055\times 25} = 8.08624 + 5.86376 = 13.95000.$$

Therefore the profit is $13.95000 - 13.93321 = 0.01679$.

(c) Calculating the convexity of the liabilities at the original interest rate,

$$conv_L := \frac{12^2 \times 10v^{12} + 15^2 \times 20v^{15}}{PV_L} = \frac{2915.93824}{14.93545} = 195.23$$

and the convexity of the assets at the original interest rate,

$$conv_A := \frac{5^2 \times X_5 + 25^2 \times X_{25}}{PV_A} = \frac{4360.09225}{14.93545} = 291.93$$

we find $conv_A > conv_L$. Hence the company is immunized in the sense of Redington, which means that small changes in the interest rate result in the new PV of the assets being greater than or equal to the new PV of the liabilities.

5. (a) As $X \sim log\mathcal{N}(\mu, \sigma^2)$

$$\text{var}(X) = \mathbb{E}\left(X^2\right) - \left(\mathbb{E}\left(X\right)\right)^2 = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1),$$

i.e.

$$\text{var}(X) = 0.04 = 1.06^2(e^{\sigma^2} - 1)$$
$$\Rightarrow \sigma^2 = \ln\left(1 + \frac{0.04}{1.06^2}\right) = 0.034981.$$

Taking square roots, $\sigma = 0.187032$. To find μ, use the expression for the mean to get

$$1.06 = e^{\mu+0.5\sigma^2} \quad \Rightarrow \quad \mu = \ln(1.06) - 0.5(0.187032)^2 = 0.040778.$$

(b) Denote by y the amount of cash to be invested now. It accumulates to yS_{10}, where

$$S_{10} = \prod_{k=1}^{10}(1 + i_k) \sim log\mathcal{N}(10\mu, 10\sigma^2).$$

We want to find y satisfying $\mathbb{P}[yS_{10} \geq 8\,000] = 0.975$.
As $\ln(S_{10}) \sim \mathcal{N}(10\mu, 10\sigma^2)$, rearrange to find

$$\begin{aligned} 0.975 =& \mathbb{P}\left[\frac{\ln(S_{10}) - 10\mu}{\sqrt{10}\sigma} \geq \frac{\ln(8\,000) - \ln(y) - 10\mu}{\sqrt{10}\sigma}\right] \\ =& 1 - \mathbb{P}\left[\frac{\ln(S_{10}) - 10\mu}{\sqrt{10}\sigma} < \frac{\ln(8\,000) - \ln(y) - 10\mu}{\sqrt{10}\sigma}\right]. \end{aligned}$$

As the random variable is standard normal, we find

$$\Phi\left(\frac{\ln(8\,000) - \ln(y) - 10\mu}{\sqrt{10}\sigma}\right) = 1 - 0.975 = 0.025.$$

Using $\Phi(-1.96) = 0.025$,

$$-1.96 = \frac{\ln(8\,000) - \ln(y) - 10\mu}{\sqrt{10}\sigma}.$$

Rearranging,

$$y = 8\,000 \exp\left(1.96\sqrt{10}\sigma - 10\mu\right) = 13\,712.55.$$

(c) The amount to invest today, £13 715.55, is greater than the amount to be paid in ten years' time, £8 000. This is because of the volatility in the rate of return; any investment in the fund may decrease in value.

(d) It does not seem appropriate, because the investor has to set aside an amount of cash (£13 715.55) today much greater than the payment to be made in ten years' time (£8 000.00), even allowing for discounting. The investor could instead invest in less risky assets, such as government-issued fixed interest bonds, and invest a much smaller amount (£8 000.00 discounted back 10 years) while still maintaining the security of being able to make the payment.

6. As $\ddot{s}^{i}_{\overline{n}|} = \frac{(1+i)^n - 1}{1-v}$,

$$\ddot{s}^{(m)}_{\overline{n}|} = \frac{1}{m}\ddot{s}^{(1+i)^{1/m}-1}_{\overline{nm}|} = \frac{(1+i)^n - 1}{m\left(1 - v^{1/m}\right)}.$$

Since $v = 1 - d = (1 - d^{(m)}/m)^m$,

$$d^{(m)} = m\left(1 - v^{1/m}\right).$$

Substituting into $\ddot{s}^{(m)}_{\overline{n}|}$, we get

$$\ddot{s}^{(m)}_{\overline{n}|} = \frac{(1+i)^n - 1}{d^{(m)}}.$$